Quiz 7

Friday, October 23 — transforms, systems, and elimination (Week 8)

Last updated: October 13, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers: Transforms (held from Week 7), plus Systems and Elimination (Week 8). RREF is on Quiz 8. Same-day Lab 2 / dynamics lecture are **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) · Lachniet Octave PDF · also on Resources.

Schedule

Core

Fluency (complete prompts — practice these by hand)

C7.1. Reflection through the x_1-axis sends $e_1 = (1,0) \mapsto (1,0)$ and $e_2 = (0,1) \mapsto (0,-1)$. Write the matrix A and compute $A(2,3)$.

C7.2. Write the 2×2 matrix for rotation by 90° counterclockwise. Where do $(1,0)$ and $(0,1)$ go?

C7.3. Let $X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$ (columns are three points) and $A = \mathrm{diag}(2,1)$. Compute AX and describe the geometric effect in one sentence.

C7.4. Predict where $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ sends the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$. Then check with a short Octave or Python computation (or a careful sketch).

C7.5. Write as $Ax = b$ and classify square / over / under:

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ -x_1 + x_3 &= 1. \end{aligned}$$

C7.6. One elimination step: clear the $(2,1)$ entry of

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 1 & 0 \end{array}\right]$$

by row replacement. Write the new matrix.

C7.7. From the echelon form

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & -3 & -6 \end{array}\right],$$

solve by back substitution.

C7.8. True or false, with a one-line reason: solving $Ax = b$ asks whether b is in the span of the columns of A.

Textbook Core

- B 7.2
- B 8.6(a), 8.9, 8.10, 8.13, 8.15
- L Ch. 2 Ex. 1: Gaussian-elimination and back-substitution portions

Stretch

- B 7.1, 7.3, 7.4, 7.8
- B 8.4(a–b), B 8.16

What a quiz item might change

Which geometric map is used, the right-hand side with the same coefficient matrix, or asking for the column interpretation of a transform versus a system $Ax = b$.